

**Department of Electronics and Telecommunication Engineering
(NBA ACCREDITED)
Digital System Design Laboratory
Academic Year 2025-2026, SEM-5, Odd Semester**

Course Code	EC24P
Subject Professor In-charge	Prof. Satendra Mane
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Roll Number	23104B0003
Class	T.E – EXTC/EXCS SEM-V
Division	B
Date of Performance	5 th September 2025
Date of Submission	12 th September 2025

EXPERIMENT NO.6

Estimated time to complete this experiment: 02 hours.

Objective: To design and implement UPCOUNTER

CO to be achieved: CO1, CO2

Expected Outcome of Experiment:

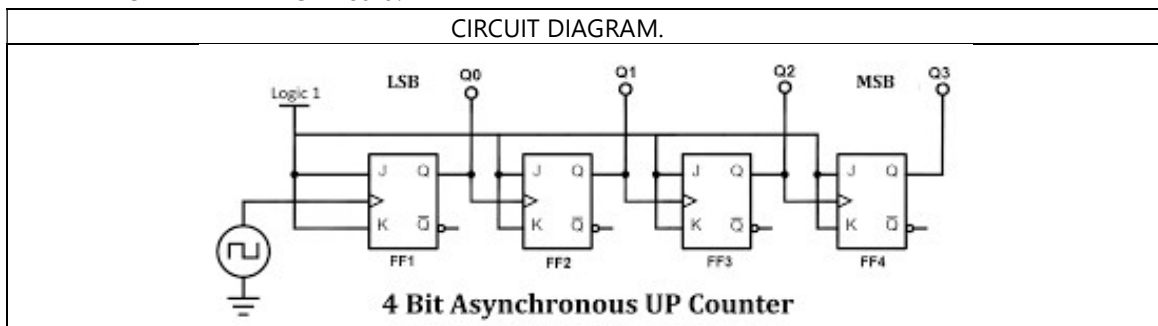
- Students will learn to design, simulate and synthesis the arithmetic circuit using VIVADO & VERILOG.

Pre Lab/ Prior Concept: Designing of Binary adders.

Apparatus/TOOL:

- PC
- VIVADO TOOL
- SPARTAN-7 FPGA Board.

CIRCUIT DIAGRAM.



VERILOG -CODE.

```
module up_counter(
    input wire clk,
    input wire reset,
    output reg [3:0] count
);
```

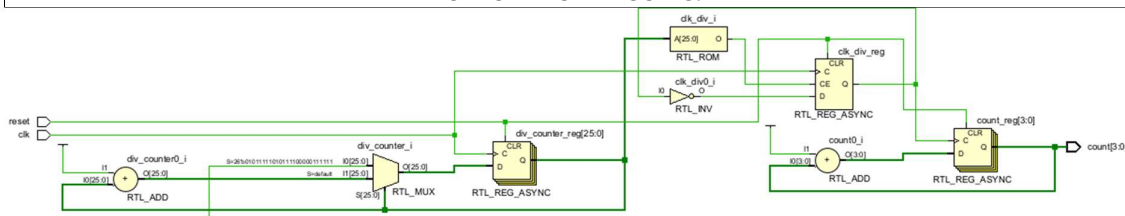
```
reg [25:0] div_counter = 0; // Counter for generating the clock division
reg clk_div = 0;           // Divided clock signal
```

```
// Clock divider logic
always @(posedge clk or posedge reset) begin
    if (reset) begin
        div_counter <= 0;
        clk_div <= 0;
    end
    else begin
        if (div_counter == 25_000_000 - 1) begin
            div_counter <= 0;
            clk_div <= ~clk_div; // Toggle the clock signal after every 25 million cycles
        end
        else begin
            div_counter <= div_counter + 1; // Increment the counter
        end
    end
end
```

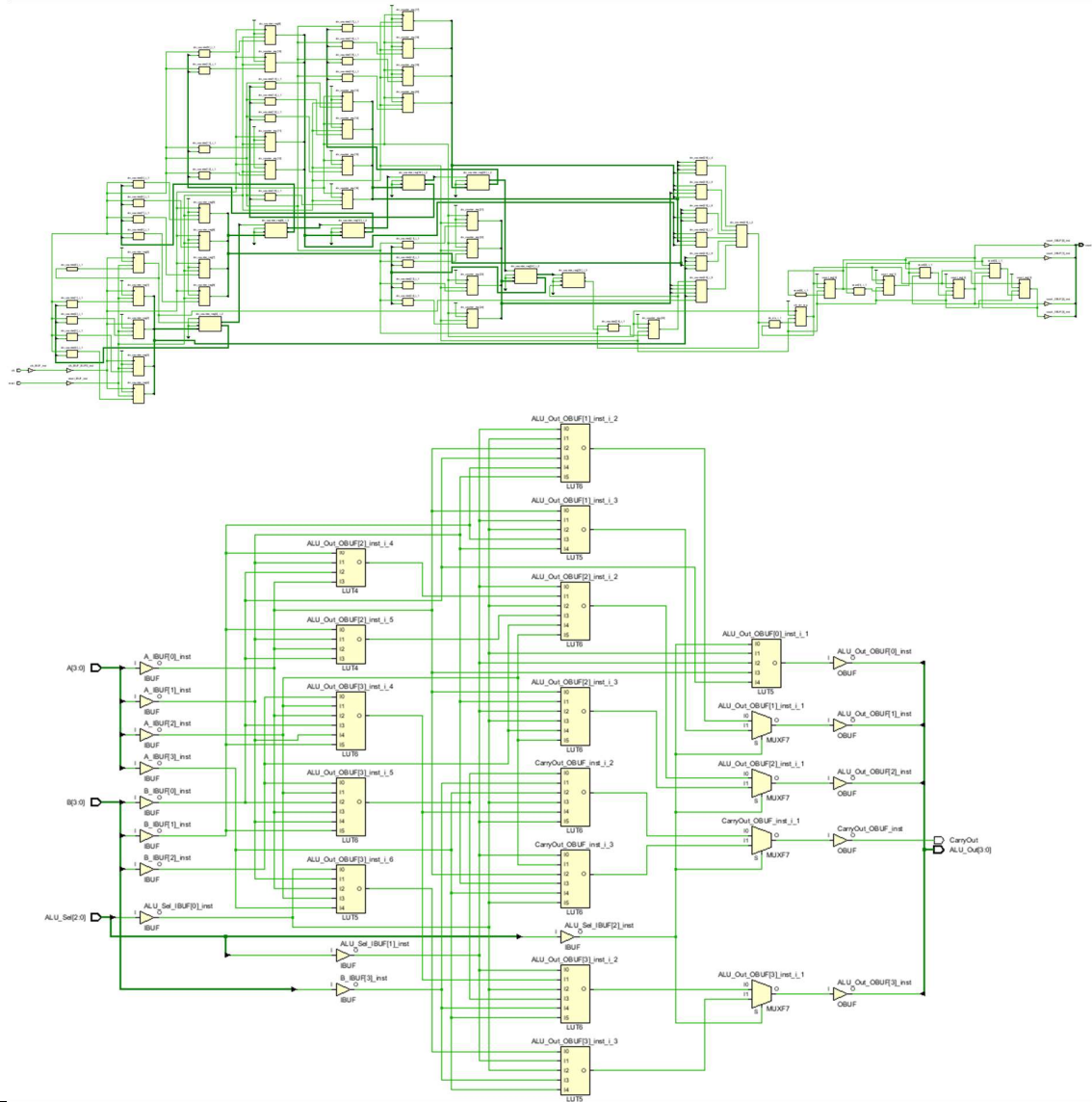
```
// Counter logic, incremented by the divided clock
always @(posedge clk_div or posedge reset) begin
    if (reset) begin
        count <= 0; // Reset the count if reset is active
    end
    else begin
        count <= count + 1; // Increment the count at each clock pulse
    end
end
```

endmodule

SIMULATION RESULTS.



SYNTHESIS MODEL.



CONCLUSION.

The 4-bit ALU was successfully designed and implemented using Verilog in Vivado. It performs arithmetic (addition, subtraction, increment, decrement) and logical (AND, OR, XOR, NOT) operations correctly. The simulation verified its functionality, and the experiment helped understand ALU design, simulation, and synthesis on FPGA

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ASSESSMENT RECORD.

Practical Performance [Model/Code, Results] (5 Marks)	Presentation/ Learning Outcome [Observation, Conclusion] (5 Marks)	Total (10 Marks)	Sign